
CS246: Database Management Systems Lab

Week # 04 (1 Questions, 75 Marks)

Lab session: AL1

Held on: 27-Jan-2025 (Mon)

Lab Timings: 14:00 to 17:00 Hours Pages: 6

Submission time: 16:45 Hrs, 27-Jan-2025

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Important information:

1. Some of the tasks are so designed to emit errors when SQL statements are invoked. You should make an informed effort to find the cause of the errors. You do not need to resolve the errors.
2. Write all the SQL statements in a text file named with `your-roll-number.sql` file name & extension and upload. Replace the text `your-roll-number` with appropriate roll number.
3. Refer to table creation document
4. Refer to table modification document
5. Refer to insert statement
6. Refer to update statement
7. Refer to delete statement

Question 1: (75 points)

Basic SQL, Chapter 3, text book *Database System Concepts*, Abraham Silberschatz, Henry F. Korth & S. Sudarshan.

Task 01 (1 mark) Create a database named `week04`. Within this database, create the following tables.

Task 02 (5 marks) **no constraints** Create a table T_1 with following columns

1. (1 mark) 1st column is of fixed length `char` data type of length 3 denoting origin
2. (1 mark) 2nd column is of fixed length `char` data type of length 3 denoting destination
3. (1 mark) 3rd column is of fixed length `char` data type of length 6 denoting flight number
4. (1 mark) 4th column is of fixed length `char` data type of length 5 denoting departure time from origin airport
5. (1 mark) 5th column is of fixed length `char` data type of length 5 denoting arrival time at destination airport

Task 03 (6 marks) Perform the following on table T_1

1. (1 mark) Insert a record having values
origin BOM
destination HYD
flight number 6E5056
departure time 21:30
arrival time 22:55
2. (1 mark) Insert a record having values
origin NAG
destination BOM
flight number 6E5302
departure time 22:55
arrival time 00:30
3. (1 mark) Insert a record having values
origin MAA
destination BOM
flight number 6E5367
departure time 19:00
arrival time 21:10
4. (1 mark) Update the record with flight number 6E5367 origin to be BLR and destination to be MAA
5. (1 mark) Update the record with flight number 6E5367 origin to be MAA and destination to be MAA
6. (1 mark) Delete the record with flight number 6E5367

Task 04 (5 marks) **primary key constraints** Create a table T_2 with following columns

1. (1 mark) 1st column is of fixed length **char** data type of length 3 denoting origin
2. (1 mark) 2nd column is of fixed length **char** data type of length 3 denoting destination
3. (1 mark) 3rd column is of fixed length **char** data type of length 6 denoting flight number. Place **primary key constraint**
4. (1 mark) 4th column is of fixed length **char** data type of length 5 denoting departure time from origin airport
5. (1 mark) 5th column is of fixed length **char** data type of length 5 denoting arrival time at destination airport

Task 05 (5 marks) Perform the following on table T_2

1. (1 mark) Insert a record having values
origin BOM
destination HYD
flight number 6E5056
departure time 21:30
arrival time 22:55

2. (1 mark) Insert a record having values

origin	NAG
destination	BOM
flight number	6E5056
departure time	22:55
arrival time	00:30
3. (1 mark) Insert a record having values

origin	MAA
destination	BOM
flight number	6E5367
departure time	19:00
arrival time	21:10
4. (1 mark) Update the record with flight number 6E5367 to flight number 6E5056.
5. (1 mark) Delete the record with flight number 6E5056

Task 06 (5 marks) **Unique key constraint & not NULL constraint** Create a table T_3 with following columns

1. (1 mark) 1st column is of fixed length **char** data type of length 3 denoting origin. Constraint as not null
2. (1 mark) 2nd column is of fixed length **char** data type of length 3 denoting destination. Constraint as not null
3. (1 mark) 3rd column is of fixed length **char** data type of length 6 denoting flight number. **Place unique and not null constraint**
4. (1 mark) 4th column is of fixed length **char** data type of length 5 denoting departure time from origin airport. Constraint as not null
5. (1 mark) 5th column is of fixed length **char** data type of length 5 denoting arrival time at destination airport. Constraint as not null

Task 07 (5 marks) Perform the following on table T_3

1. (1 mark) Insert a record having values

origin	NULL
destination	HYD
flight number	6E5056
departure time	21:30
arrival time	22:55
2. (1 mark) Insert a record having values

origin	BOM
destination	NULL
flight number	6E5056
departure time	21:30
arrival time	22:55
3. (1 mark) Insert a record having values

origin	BOM
destination	HYD
flight number	NULL
departure time	21:30
arrival time	22:55

4. (1 mark) Insert a record having values

origin	BOM
destination	HYD
flight number	6E5056
departure time	NULL
arrival time	22:55

5. (1 mark) Insert a record having values

origin	BOM
destination	HYD
flight number	6E5056
departure time	21:30
arrival time	NULL

Task 08 (5 marks) **Data Quality issues** Create a table T_4 with following columns

- (1 mark) 1st column is of fixed length **char** data type of length 3 denoting origin. Constraint as not null
- (1 mark) 2nd column is of fixed length **char** data type of length 3 denoting destination. Constraint as not null
- (1 mark) 3rd column is of fixed length **char** data type of length 6 denoting flight number. **Place primary key constraint**
- (1 mark) 4th column is of fixed length **char** data type of length 5 denoting departure time from origin airport. Constraint as not null
- (1 mark) 5th column is of fixed length **char** data type of length 5 denoting arrival time at destination airport. Constraint as not null

Task 09 (5 marks) Perform the following on table T_4

1. (1 mark) Insert a record having values

origin	BOM
destination	BOM
flight number	6E5056
departure time	21:30
arrival time	22:55

2. (1 mark) Insert a record having values

origin	BOM
destination	HYD
flight number	6E5056
departure time	22:55
arrival time	21:30

3. (1 mark) Insert a record having values

origin	BOM
destination	HYD
flight number	6E
departure time	21:30
arrival time	22:55

4. (1 mark) Insert a record having values

origin	MOB
destination	DYH
flight number	6E5056
departure time	2255:
arrival time	2130:

5. (1 mark) Insert a record having values

origin	MOBOM
destination	DYHYD
flight number	6E5056-6E5056
departure time	21:30
arrival time	22:55

Task 10 (5 marks) **Good table** Create a table T_5 with following columns

- (1 mark) 1st column is of fixed length **char** data type of length 3 denoting origin.
- (1 mark) 2nd column is of fixed length **char** data type of length 3 denoting destination.
- (1 mark) 3rd column is of fixed length **char** data type of length 6 denoting flight number. **Place primary key constraint**
- (1 mark) 4th column is of fixed length **char** data type of length 5 denoting departure time from origin airport.
- (1 mark) 5th column is of fixed length **char** data type of length 5 denoting arrival time at destination airport.

Task 11 (4 marks) Perform the following on table T_5

- (1 mark) Insert a record having values

origin	BOM
flight number	6E5056
- (1 mark) Update the record having flight number 6E5056 destination value HYD
- (1 mark) Update the record having flight number 6E5056 departure time 21:30
- (1 mark) Update the record having flight number 6E5056 arrival time 22:55

Task 12 (1 mark) **Altering table** Create a table T_7 with following columns

- (1 mark) 3rd column is of fixed length **char** data type of length 6 denoting flight number.

Task 13 (23 marks) Perform the following on table T_7

- (1 mark) Insert a record having value flight number 6E5056

2. (1 mark) Insert a record having value flight number 6E5302
3. (1 mark) Insert a record having value flight number 6E5367
4. (2 marks) Alter table and add column origin of fixed length `char` data type having length 3.
5. (1 mark) Update the origin of flight number 6E5056 to BOM
6. (1 mark) Update the origin of flight number 6E5302 to NAG
7. (1 mark) Update the origin of flight number 6E5367 to MAA
8. (2 marks) Alter table and add column destination of fixed length `char` data type having length 3.
9. (1 mark) Update the destination of flight number 6E5056 to HYD
10. (1 mark) Update the destination of flight number 6E5302 to BOM
11. (1 mark) Update the destination of flight number 6E5367 to BOM
12. (2 marks) Alter table and add column departure of fixed length `char` data type having length 5.
13. (1 mark) Update the departure time of flight number 6E5056 to 21:30
14. (1 mark) Update the departure time of flight number 6E5302 to 22:55
15. (1 mark) Update the departure time of flight number 6E5367 to 19:00
16. (2 marks) Alter table and add column arrival of fixed length `char` data type having length 5.
17. (1 mark) Update the arrival time of flight number 6E5056 to 22:55
18. (1 mark) Update the arrival time of flight number 6E5302 to 00:30
19. (1 mark) Update the arrival time of flight number 6E5367 to 21:10